2. Newtonian evolution of Cosmological Perturb.

Matter as an ideal, non-relativistic self-gravit fluid. Consider the epoch at which this fluid is the dominant component of energy density in the Universe.

Dynamical equations:

$$\frac{\partial \rho}{\partial t}[\vec{r},t] + \frac{\partial}{\partial \vec{r}}(\rho \vec{u})[\vec{r},t] = 0$$

$\vec{r}$ is the position of fluid element in physical (proper) coordinates, $\vec{v}$ its corresponding velocity.

Conservation of momentum (Euler's equations)

$$\frac{\partial \vec{u}}{\partial t}[\vec{r},t] + \left(\vec{u}\frac{\partial}{\partial \vec{r}}\right)\vec{u}[\vec{r},t] = -\frac{\partial}{\partial \vec{r}}\psi[\vec{r},t] - \frac{1}{\rho}\frac{\partial}{\partial \vec{r}}P[\vec{r},t]$$

ψ is the gravitational potential, which obeys the Poisson equation:

$$\left(\frac{\partial}{\partial \vec{r}}\right)^2 \psi[\vec{r},t] = 4\pi G \rho[\vec{r},t]$$

Change to comoving coordinates: $\vec{x} \equiv \frac{\vec{r}}{a(t)}$

$\vec{u} = \dot{a}\vec{x} + \vec{v}[\vec{x},t]$

Hubble flow peculiar velocity

- $\frac{\partial \rho}{\partial t}[\vec{x},t] + 3\frac{\dot{a}}{a}\rho[\vec{x},t] + \frac{1}{a}\frac{\partial}{\partial \vec{x}}(\rho\vec{v})[\vec{x},t] = 0$
- $\frac{\partial v}{\partial t}[\vec{x},t] + \frac{1}{a}\left(\vec{v}\frac{\partial}{\partial \vec{x}}\right)\vec{v}[\vec{x},t] + \frac{\dot{a}}{a}\vec{v}[\vec{x},t]$
$= -\frac{1}{a}\frac{\partial}{\partial \vec{x}}\phi[\vec{x},t] - \frac{1}{a\rho}\frac{\partial}{\partial \vec{x}}P[\vec{x},t]$
- $\left(\frac{1}{a}\frac{\partial}{\partial \vec{x}}\right)^2 \phi[\vec{x},t] = 4\pi G\rho[\vec{x},t] + 3\frac{\ddot{a}}{a}$

$\phi = \psi + \frac{1}{2}a\ddot{a}\vec{x}^2$

Consider case where fluid is only slightly perturbed

- $\rho[\vec{x},t] = \bar{\rho}(t)(1+\delta[\vec{x},t])$ where

$\delta[\vec{x},t] \equiv \frac{\rho[\vec{x},t] - \bar{\rho}(t)}{\bar{\rho}(t)}$ is density contrast $\ll 1$

- $\vec{v}[\vec{x},t] \ll \dot{a}\vec{x}$
- $P[\vec{x},t] = \bar{P}[t] + \delta P[\vec{x},t] = \bar{P}[t] + \left(\frac{\partial P}{\partial \rho}\right)_S \delta\rho[\vec{x},t]$
$= \bar{P}[t] + c_s^2\bar{\rho}\delta[\vec{x},t]$

for adiabatic perturbation with associated sound speed c_s.

- $\phi[\vec{x},t] \ll \frac{1}{2} a\ddot{a}\vec{x}^2$

Linearise the previous equations:

- $\frac{\partial \delta}{\partial t}[\vec{x},t] + \frac{1}{a}\frac{\partial \vec{v}}{\partial \vec{x}}[\vec{x},t] = 0$
- $\frac{\partial \vec{v}}{\partial t}[\vec{x},t] + \frac{\dot{a}}{a} v[\vec{x},t] = -\frac{1}{a}\frac{\partial}{\partial \vec{x}}\phi[\vec{x},t] - \frac{c_s^2}{a}\frac{\partial}{\partial \vec{x}}\delta[\vec{x},t]$
- $\nabla_{\vec{x}}^2 \phi[\vec{x},t] = 4\pi G \bar{\rho} a^2 \delta[\vec{x},t]$

NB. $\left(\frac{\partial}{\partial \vec{r}}\right)_t = \frac{1}{a}\left(\frac{\partial}{\partial \vec{x}}\right)_t \qquad \left(\frac{\partial}{\partial t}\right)_{\vec{r}} = \left(\frac{\partial}{\partial t}\right)_{\vec{x}} - \frac{\dot{a}}{a}\left(\vec{x}\cdot\frac{\partial}{\partial \vec{x}}\right)_t$

- FL eq in 1st lecture to get rid of 0th order terms -

$\nabla_{\vec{x}}$ (Euler eq) and use continuity and Poisson to get rid of $\nabla_{\vec{x}}\vec{v}$ and $\nabla_x^2\phi$.

$$\ddot{\delta}[\vec{x},t] + 2H\dot{\delta}[\vec{x},t] - \frac{c_s^2}{a^2}\nabla_{\vec{x}}^2\delta[\vec{x},t] = 4\pi G\bar{\rho}\delta[\vec{x},t]$$

Fourier transform of

$$\delta[\vec{x},t] \equiv \frac{1}{(2\pi)^3}\int d^3k\, \delta_k \exp(-i\vec{k}\cdot\vec{x})$$

$$\ddot{\delta}_{\vec{k}} + 2H\dot{\delta}_{\vec{k}} + \left(\frac{c_s^2}{a^2}k^2 - 4\pi G\bar{\rho}\right)\delta_{\vec{k}} = 0$$

using the conformal version of it $d\eta \equiv \frac{dt}{a}$

$\cdot \longrightarrow '$

$\frac{\partial}{\partial t} \longrightarrow \frac{\partial}{\partial \eta}$

Conformal Hubble parameter:

$$\mathcal{H} \equiv \frac{a'}{a} = aH$$

$$\delta''_{\vec{k}} + \mathcal{H}\delta'_{\vec{k}} + \left(c_s^2 k^2 - 4\pi G a^2 \bar{\rho}\right)\delta_{\vec{k}} = 0$$